Course: Enterprise Real-Time Data Engineering
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Course Portal: fakhr.me/KafkaSpark
Code & Labs Repository: github.com/fakhruldeen/KafkaSpark

Troubleshooting Runbook: Local Lab Environment

Diagnostic Solutions for Trainees Working from Home or Office Laptops

Course: Apache Kafka Core, Confluent Developer & Apache Spark

1. Top Issues & Quick Fixes

Issue 1: "Connection Refused" or `TimeoutException`

- **Symptom:** You run a Python script or CLI command and it hangs for 60 seconds, then crashes with `TimeoutException` or `Connection refused`.
- **The Root Cause:** Network context mismatch.
- **The Rule:**
- **From your laptop terminal:** Connect to `localhost:9092`.
- **From inside a Docker container (e.g. `docker exec`):** Connect to `kafka1:9092`.
- *Never type `localhost:9092` inside a container! Inside a container, `localhost` means the container itself, which is not listening on that port.*

Issue 2: Port Conflict (`port is already allocated` / `address already in use`)

- **Symptom:** `docker compose up` fails with:
  `Bind for 0.0.0.0:5432 failed: port is already allocated` or port `8080` / `9092`.
- **The Root Cause:** You have an existing local installation of PostgreSQL, Jenkins, Tomcat, or an older Kafka instance running on your laptop.
- **The Fix:**
  Find and stop the occupying service:
  ```bash
  # Check what process is using port 5432 (Postgres) or 8080 (Web)
  ```

sudo lsof -i :5432
sudo lsof -i :8080

On Ubuntu, stop system PostgreSQL if running:
sudo systemctl stop postgresql
~~~

Issue 3: Docker Out of Memory / Containers Exit with Code 137

- **Symptom:** A container stops unexpectedly. `docker ps -a` shows `Exited (137)`.
- **The Root Cause:** Exit code 137 is the Linux kernel OOM Killer (Out Of Memory). The workstation did not allocate enough RAM to Docker.
- **The Fix:**
- **Windows / macOS Docker Desktop:**
  Open Docker Desktop ⟶ Settings ⟶ Resources ⟶ Set Memory slider to **at least 8 GB** (10 GB recommended).
- **Windows WSL 2 `.wslconfig`:**
  Create a file named `C:\Users\<YourUsername>\.wslconfig` with:
  ```ini
  [wsl2]
  memory=10GB
  processors=4
  ```
  Then restart WSL in PowerShell: `wsl --shutdown`.

Issue 4: Docker Permission Denied (`permission denied while trying to connect to the Docker daemon socket`)

- **Symptom:** Running `docker ps` outputs `permission denied`.
- **The Fix:** Add your Linux user to the `docker` group:
  ```bash
  sudo usermod -aG docker $USER
  newgrp docker
  ```

Issue 5: PySpark JVM Initialization Error (`Py4JJavaError` / `JAVA_HOME is not set`)

- **Symptom:** Running `python -c "import pyspark"` throws `Java not found` or `JAVA_HOME is not set`.
- **The Fix:**
  1. Confirm OpenJDK 17 is installed:
  ```bash
  ```
~~~

```
    sudo apt install -y openjdk-17-jdk-headless
```

2. Set `JAVA_HOME` in your `~/.bashrc`:

```
bash
    echo 'export JAVA_HOME=/usr/lib/jvm/java-17-openjdk-amd64' >> ~/.bashrc
    echo 'export PATH=←!—KATEX_INLINE_3—→PATH' >> ~/.bashrc
    source ~/.bashrc
```

Issue 6: Cleaning and Resetting the Entire Stack (The Nuclear Option)

If your topics, offsets, or schemas get hopelessly corrupted:

```
cd 00_Lab

# Stop all containers and wipe all lab volumes cleanly
make reset

# Start fresh
make up
make check
```

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